# Counting and probability solutions

## C-E01 Two offices versus an unordered group

### Derivation and check

There are $7 \cdot 6 = 42$ unrestricted assignments. If $A$ is deputy, the president can be any of the other six people. Excluding these six assignments leaves $36$.

Six unordered pairs contain $A$, and each has just one valid assignment. The other $\binom{6}{2} = 15$ pairs each have two valid assignments. Thus $6 + 2 \cdot 15 = 36$, but the overcounting is not uniform, so $36/2$ is not the group count. There are $6 + 15 = 21 = \binom{7}{2}$ groups with at least one valid assignment. Allowing one person to hold both offices would change the sample space.

## C-E02 String positions and repeated keys

### Derivation and check

The length-three substrings starting at positions $1, 2, 3$ are `aba`, `bab`, and `aba`. There are three positional substrings and two distinct contents.

A positional subsequence chooses whether to include each of the five positions. There are $2^5 = 32$ choices, including the empty choice. The distinct-content count is smaller. Even the single-character string `a` has three possible position choices.

The number of key strings is

$$\frac{5!}{2!2!1!} = 30.$$

Each key string has $2!$ permutations of the `a` identities and $2!$ permutations of the `b` identities, hence exactly four identity permutations. The divisor is valid because this overcounting is uniform. Equal content and equal position choice are different counting objects.

## C-E03 Binomial coefficients and entropy

### Derivation and check

Partition the $k$-subsets according to whether they omit or contain a specified element. The two class sizes give Pascal's identity:

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

For $0 \le j < k \le n$, each factor $(n-j)/(k-j) \ge n/k$, proving the lower product bound. Bounding the numerator by $n^k$ and using $k! \ge (k/e)^k$ proves the upper bound:

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$$\left(\frac{n}{k}\right)^k \le \binom{n}{k} \le \frac{n^k}{k!} \le \left(\frac{en}{k}\right)^k.$$

For $\alpha = k/n \in (0,1)$, the nonnegative expansion

$$\sum_{j=0}^{n} \binom{n}{j} \alpha^j (1-\alpha)^{n-j} = 1$$

has a $k$th term at most $1$. Dividing by its positive probability factors gives

$$\binom{n}{k} \le \alpha^{-k}(1-\alpha)^{-(n-k)} = 2^{nH(\alpha)}.$$

At $n = 8, k = 2$, the true value is $28$. The lower bound is $16$, the coarse upper bound is $16e^2 \approx 118.2249$, and the entropy upper bound is $4^2(4/3)^6 \approx 89.8985$. All are valid but loose. The bound $\binom{n}{k} \le 2^n$ is another option; no one bound is best for all parameters. Endpoints use empty products or continuous extension.

## C-E04 Conditional sampling

### Derivation and check

The probabilities are

$$\begin{aligned} P(\text{both green}) &= \frac{4}{7}\frac{3}{6} = \frac{2}{7}, \\ P(\text{second green}) &= \frac{4}{7}\frac{3}{6} + \frac{3}{7}\frac{4}{6} = \frac{4}{7}, \\ P(\text{second green} \mid \text{first black}) &= \frac{4}{6} = \frac{2}{3}. \end{aligned}$$

Independence would give $(4/7)^2 = 16/49$ for both green, whereas the actual value is $2/7 = 14/49$. Thus independence fails.

In general, disjoint events with positive probabilities satisfy $P(A \cap B) = 0$ while $P(A)P(B) > 0$, so they are dependent. Equal first- and second-draw marginals do not imply independence. Conditioning reflects the change caused by sampling without replacement.

## C-E05 Pairwise is not mutual independence

### Derivation and check

The outcomes $000, 011, 101, 110$ each have probability $1/4$. Every pair $(U,V)$, $(U,W)$, and $(V,W)$ sees $00, 01, 10, 11$ once, so every pairwise joint mass equals $1/4$ and factors into its marginals. Yet

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$$P(U = V = W = 1) = 0 \neq \frac{1}{8} = P(U = 1)P(V = 1)P(W = 1).$$

Each bit has variance $1/4$, and every pair has covariance zero. Variance addition therefore gives $\operatorname{Var}(U + V + W) = 3/4$.

For a direct check, the sum is $0$ with probability $1/4$ and $2$ with probability $3/4$. Its mean is $3/2$, its second moment is $3$, and its variance is $3 - 9/4 = 3/4$. Mean linearity needs no independence. Variance addition needs zero pairwise covariances. Pairwise independence alone cannot factor triple intersections or products.

## C-E06 Bayes and observation rules

### Derivation and check

For one alarm,

$$P(\text{alarm}) = 0.01 \cdot 0.8 + 0.99 \cdot 0.05 = 0.0575,$$
$$P(\text{fault} \mid \text{alarm}) = \frac{0.008}{0.0575} = \frac{16}{115} \approx 0.139130.$$

Conditional independence gives

$$P(\text{two alarms}) = 0.01 \cdot 0.8^2 + 0.99 \cdot 0.05^2 = 0.008875,$$
$$P(\text{fault} \mid \text{two alarms}) = \frac{0.0064}{0.008875} = \frac{256}{355} \approx 0.721127.$$

If both tests use exactly the same output, two alarms are the same event as one alarm, and the posterior remains $16/115$. The individual test marginals still match. Without a conditional-dependence model, the two-test answer is undetermined.

## C-E07 Joint masses, mean, and variance

### Derivation and check

For $m = 1, 2, 3$, the maximum has mass $(2m - 1)/9$, giving $1/9, 3/9, 5/9$. Its moments and tail are

$$E[M] = \frac{1 + 6 + 15}{9} = \frac{22}{9},$$
$$E[M^2] = \frac{1 + 12 + 45}{9} = \frac{58}{9},$$
$$\operatorname{Var}(M) = \frac{58}{9} - \frac{484}{81} = \frac{38}{81},$$
$$P(M \geq 2) = \frac{8}{9}.$$

For the uniform die $A$, $\operatorname{Var}(A) = 2/3$. With $X = A, Y = 4 - A$,

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$$\operatorname{Cov}(X, Y) = -\frac{2}{3}, \qquad X + Y \equiv 4.$$

Thus the sum variance is zero. Adding only the individual variances gives $4/3$ and misses $2\operatorname{Cov}(X, Y) = -4/3$. Dependence can increase fluctuations or cancel them completely.

## C-E08 Tail sums and nonlinear expectation

### Derivation and check

The mass-based mean is $0 \cdot (1/2) + 1 \cdot (1/4) + 3 \cdot (1/4) = 1$. The tails at thresholds $1, 2, 3$ have probabilities $1/2, 1/4, 1/4$, and all higher tails vanish. Their sum also equals $1$.

The second moment is $E[X^2] = 1/4 + 9/4 = 5/2$, and the variance is $3/2$. Jensen for the square function gives $E[X^2] = 2.5 \geq (E[X])^2 = 1$.

For the constant variable $X \equiv -1$, the true mean is $-1$, but $\sum_{i \geq 1} P(X \geq i) = 0$. The tail formula requires nonnegative integer support. Signed variables may be split into positive and negative parts, provided undefined expressions such as $\infty - \infty$ are avoided.

## C-E09 Two geometric counting conventions

### Derivation and check

The two mass functions are

$$P(G = k) = \left(\frac{3}{5}\right)^{k-1} \frac{2}{5}, \quad k \geq 1,$$

$$P(F = f) = \left(\frac{3}{5}\right)^{f} \frac{2}{5}, \qquad f \geq 0.$$

Their means are $E[G] = 5/2$ and $E[F] = 3/2$, and both have variance $15/4$. The tails are

$$P(G > 4) = \left(\frac{3}{5}\right)^{4} = \frac{81}{625},$$

$$P(G > 7 \mid G > 3) = \frac{q^7}{q^3} = q^4 = \frac{81}{625}.$$

The total time is $2F + 5$ ms, with expectation $2 \cdot 1.5 + 5 = 8$ ms. At $p = 0$, success never occurs and the trial count is infinite, outside the finite-valued geometric mass function. At $p = 1$, $G = 1, F = 0$, the variance is zero, and the time is $5$ ms. The trial mean is not the failure mean.

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# Counting and probability solutions (continued)

## C-E10 Binomial table and mode

### Derivation and check

For $k = 0, \ldots, 6$, the numerators $\binom{6}{k}2^{6-k}$ are

$$64, \quad 192, \quad 240, \quad 160, \quad 60, \quad 12, \quad 1.$$

Their sum is $729$. The first weighted sum is $1458$, giving $E[X] = 2$. The second weighted sum is $3888$, giving $E[X^2] = 16/3$ and variance $16/3 - 4 = 4/3 = npq$. The tail is

$$P(X \geq 4) = \frac{60 + 12 + 1}{729} = \frac{73}{729}.$$

The adjacent ratio is $b(k)/b(k-1) = (7-k)/(2k)$. It exceeds $1$ exactly when $k < 7/3$, so the unique mode is $2$.

For a shared trial, $X = 0$ with probability $2/3$ and $X = 6$ with probability $1/3$. The mean remains $2$, but the variance is $36 \cdot 2/9 = 8$. The binomial mass formula no longer applies.

## C-E11 Union and geometric tail bounds

### Derivation and check

The exact tail is $1789/65536$. The union bound gives

$$\binom{8}{5}\left(\frac{1}{4}\right)^5 = \frac{7}{128} = \frac{3584}{65536}.$$

At $k = 4$, the right-ratio coefficient is $(8-4)(1/4)/(4-2) = 1/2$, giving the bound $2835/65536$.

For the general left-tail proof, assume $0 < p < 1$ and an integer $0 < k < np$. For $1 \leq i \leq k$,

$$\frac{b(i-1)}{b(i)} = \frac{iq}{(n-i+1)p} < \rho = \frac{kq}{(n-k)p} < 1.$$

Working backward gives $b(k-j) < \rho^j b(k)$ for $j = 1, \ldots, k$. Therefore

$$\begin{aligned} P(X < k) &= \sum_{i=0}^{k-1} b(i) \\ &< b(k) \sum_{j \geq 1} \rho^j = \frac{kq}{np-k} b(k). \end{aligned}$$

The exact tail is below the ratio bound, which is below the union bound in this example; this ordering is not universal. The event $X < k$ excludes $b(k)$, and the event $X > k$ also excludes $b(k)$. That mass is only the reference term controlling each tail.

# Counting and probability solutions (continued)

## C-E12 Exponential bounds and quantifiers

### Derivation and check

For $\lambda > 0$, Markov and mutual independence give

$$\begin{aligned} P(X \geq t) &\leq e^{-\lambda t} E[e^{\lambda X}] \\ &= e^{-\lambda t} \prod_i E[e^{\lambda I_i}]. \end{aligned}$$

Each factor satisfies

$$E[e^{\lambda I_i}] = 1 + p_i(e^\lambda - 1) \leq \exp[p_i(e^\lambda - 1)].$$

Thus the bound is $\exp[\mu(e^\lambda - 1) - \lambda t]$. For $t > \mu > 0$, the exponent has derivative $\mu e^\lambda - t$ and positive second derivative $\mu e^\lambda$. Its minimum is at $\lambda = \ln(t/\mu) > 0$. Substitution gives

$$P(X \geq t) \leq e^{-\mu} \left(\frac{e\mu}{t}\right)^t.$$

For the centered form,

$$\begin{aligned} E[e^{\lambda(I_i - p_i)}] &= p_i e^{\lambda(1-p_i)} + (1-p_i)e^{-\lambda p_i} \\ &\leq 1 + p_i e^\lambda \leq \exp(p_i e^\lambda). \end{aligned}$$

Factoring centered moments and applying Markov gives

$$P(X - \mu \geq r) \leq \exp(\mu e^\lambda - \lambda r).$$

For $r > \mu > 0$, minimize at $\lambda = \ln(r/\mu) > 0$ to obtain

$$P(X - \mu \geq r) \leq \left(\frac{e\mu}{r}\right)^r.$$

This centered event has total threshold $\mu + r$.

For $\mu = 10$, the event $X \geq 50$ uses $r = 40$. Markov gives $0.2$. The true variance $9.9$ gives the Chebyshev bound $0.0061875$. The centered exponential bound is approximately $1.947 \times 10^{-7}$, and the refined total-threshold bound is approximately $2.650 \times 10^{-18}$. Both exponential bounds require mutual independence. Chebyshev requires the correct variance.

If $\mu = 0$, nonnegative $p_i$ force every $p_i = 0$. Thus $X = 0$ almost surely, and every positive-threshold tail is zero. Do not substitute into a logarithm involving division by zero.

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# Counting and probability solutions (continued)

## Verification code appendix: probability_checks.py

```
"""Exact small distributions plus numerical log-tail audit; not proofs of bounds."""
from fractions import Fraction as F
from math import comb,exp,log,lgamma
from itertools import product
def pmf(n,p):return [F(comb(n,k))*p**k*(1-p)**(n-k) for k in range(n+1)]
if __name__=='__main__':
 for n in range(1,21):
  for p in [F(1,4),F(1,3),F(1,2),F(2,3)]:
   b=pmf(n,p);mu=n*p;q=1-p
   assert sum(b)==1 and sum(k*b[k] for k in range(n+1))==mu
   assert sum(k*k*b[k] for k in range(n+1))-mu*mu==n*p*q
   for k in range(n+1):
    assert sum(b[k:])<=comb(n,k)*p**k
    if 0<k<mu:assert sum(b[:k])<k*q/(mu-k)*b[k]
    if mu<k<n:assert sum(b[k+1:])<(n-k)*p/(k-mu)*b[k]
 b=pmf(8,F(1,4));assert [x*65536 for x in b]==[6561,17496,20412,13608,5670,1512,252,24,1]
 assert sum(b[5:])==F(1789,65536)
 b=pmf(6,F(1,3));assert [x*729 for x in b]==[64,192,240,160,60,12,1]
 assert F(8,1000)/(F(8,1000)+F(99,100)*F(5,100))==F(16,115)
 pairs=[(u,v,u^v) for u,v in product(range(2),repeat=2)]
 for i,j in [(0,1),(0,2),(1,2)]:assert len({(s[i],s[j]) for s in pairs})==4
 logs=[lgamma(1001)-lgamma(k+1)-lgamma(1001-k)+k*log(.01)+(1000-k)*log(.99) for k in
range(50,1001)]
 m=max(logs);tail=exp(m)*sum(exp(z-m) for z in logs)
 weak=exp(40*(1+log(10/40)));sharp=exp(-10+50*(1+log(10/50)))
 assert tail<sharp<weak<9.9/1600<.2
 print('Bin1000 p.01 P>=50',tail,'weak',weak,'sharp',sharp)
 print('PASS finite80 binomial models exact moments/union/ratio tails; exact Bayes and
pairwise table; stable log-sum tail audit')
```

### References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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